tf.random.uniform

View source on GitHub (https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/random_ops.py#L211-L320)

Outputs random values from a uniform distribution.

```
tf.random.uniform(
    shape,
    minval=0,
    maxval=None,
    dtype=tf.dtypes.float32 (https://www.tensorflow.org/api_docs/python/tf/dtypes#float32),
    seed=None,
    name=None
)
```

Used in the notebooks

Used in the guide	Used in the tutorials
• Validating correctness & numerical equivalence (https://www.tensorflow.org/guide/migrate/validate_correctness) • tf.data: Build TensorFlow input pipelines (https://www.tensorflow.org/guide/data) • Introduction to graphs and tf.function (https://www.tensorflow.org/guide/intro_to_graphs) • Logistic regression for binary classification with Core APIs (https://www.tensorflow.org/guide/core/logistic_regression_core) • Quickstart for the TensorFlow Core APIs (https://www.tensorflow.org/guide/core/quickstart_core)	• Customization basics: tensors and operations (https://www.tensorflow.org/tutorials/customization/basics) • Parameter server training with ParameterServerStrategy (https://www.tensorflow.org/tutorials/distribute/parameter_server_training) • Learned data compression (https://www.tensorflow.org/tutorials/generative/data_compression) • DeepDream (https://www.tensorflow.org/tutorials/generative/deepdream) • pix2pix: Image-to-image translation with a conditional GAN (https://www.tensorflow.org/tutorials/generative/pix2pix)

The generated values follow a uniform distribution in the range `[minval, maxval)`. The lower bound `minval` is included in the range, while the upper bound `maxval` is excluded.

For floats, the default range is `[0, 1)`. For ints, at least `maxval` must be specified explicitly.

In the integer case, the random integers are slightly biased unless `maxval - minval` is an exact power of two. The bias is small for values of `maxval - minval` significantly smaller than the range of the output (either `2**32` or `2**64`).

Examples:

```
>>> tf.random.uniform(shape=[2])
<tf.Tensor: shape=(2,), dtype=float32, numpy=array([..., ...], dtype=float32)>
>>> tf.random.uniform(shape=[], minval=-1., maxval=0.)
<tf.Tensor: shape=(), dtype=float32, numpy=-...>
>>> tf.random.uniform(shape=[], minval=5, maxval=10, dtype=tf.int64)
<tf.Tensor: shape=(), dtype=int64, numpy=...>
```

The `seed` argument produces a deterministic sequence of tensors across multiple calls. To repeat that sequence, use `tf.random.set_seed` (https://www.tensorflow.org/api_docs/python/tf/random/set_seed):

```
>>> tf.random.set_seed(5)
>>> tf.random.uniform(shape=[], maxval=3, dtype=tf.int32, seed=10)
<tf.Tensor: shape=(), dtype=int32, numpy=2>
>>> tf.random.uniform(shape=[], maxval=3, dtype=tf.int32, seed=10)
<tf.Tensor: shape=(), dtype=int32, numpy=0>
>>> tf.random.set_seed(5)
```

```
>>> tf.random.uniform(shape=[], maxval=3, dtype=tf.int32, seed=10)
<tf.Tensor: shape=(), dtype=int32, numpy=2>
>>> tf.random.uniform(shape=[], maxval=3, dtype=tf.int32, seed=10)
<tf.Tensor: shape=(), dtype=int32, numpy=0>
```

Without `tf.random.set_seed` (https://www.tensorflow.org/api_docs/python/tf/random/set_seed) but with a `seed` argument is specified, small changes to function graphs or previously executed operations will change the returned value. See `tf.random.set_seed` (https://www.tensorflow.org/api_docs/python/tf/random/set_seed) for details.

Args	
`shape`	A 1-D integer Tensor or Python array. The shape of the output tensor.
`minval`	A Tensor or Python value of type **`dtype`**, broadcastable with **`shape`** (for integer types, broadcasting is not supported, so it needs to be a scalar). The lower bound on the range of random values to generate (inclusive). Defaults to 0.
`maxval`	A Tensor or Python value of type **`dtype`**, broadcastable with **`shape`** (for integer types, broadcasting is not supported, so it needs to be a scalar). The upper bound on the range of random values to generate (exclusive). Defaults to 1 if **`dtype`** is floating point.
`dtype`	The type of the output: **`float16`**, **`bfloat16`**, **`float32`**, **`float64`**, **`int32`**, or **`int64`**. Defaults to **`float32`**.
`seed`	A Python integer. Used in combination with `tf.random.set_seed` (https://www.tensorflow.org/api_docs/python/tf/random/set_seed) to create a reproducible sequence of tensors across multiple calls.
`name`	A name for the operation (optional).

Returns
A tensor of the specified shape filled with random uniform values.

Raises	
`ValueError`	If **`dtype`** is integral and **`maxval`** is not specified.